School of Computer Science
COMP 2067 Programming for Beginners
Summer 2025 – Assignment 3
Course instructor: Dr. Olena Syrotkina

For each question, write a program **with comments** to perform the required tasks.

Question 1.
Find the *username* that comes first alphabetically from a list of strings.
*The **username** may contain alphabetic and non-alphabetic characters (e.g., numbers or symbols in usernames), but you can assume all alphabetic characters are in lowercase (no uppercase letters).*
Note: Do NOT use the sort() method, or built-in min() or max() functions!

a. Define a list "***users***" consisting of at least 10 user names, some with alphabetic characters and some containing numbers or special characters. For example:
   `users = ['john123', 'amy', 'bob', 'charlie99', 'david', 'eva_01', 'zoe', 'mark$', 'nina', 'liam'].`
   Print the list of users.
b. Initialize a string variable ***first_user*** to an empty string. This will store the username that comes first alphabetically from valid alphabetic names checked so far.
c. Loop through each username in the list do the following:
   i. Check if the user name contains only alphabetic characters using **isalpha()**. If valid, compare it with the current ***first_user*** and update ***first_user*** if it comes earlier alphabetically.
   ii. If it contains any non-alphabetic characters, ignore it and print a message similar to this one:
   `Ignored: 'john123' contains non-alphabetic characters.`
d. After exiting the loop:
   i. If no valid alphabetic usernames were found, print an appropriate message.
   ii. Otherwise, print the username that comes first alphabetically.

Question 2.
Find the Greatest Common Divisor (GCD) of two positive integers.

a. Initialize a variable ***x*** to a positive integer between 1 and 100.
b. Ask the user to enter a positive integer between 1 and 100, assign it to variable ***y***. Assume the user will enter valid input in the proper range.
c. Find the smaller of the two integers and store it in a separate variable ***z***.
d. In a loop, do the following:

   i. Check if both *x* and *y* are divisible by *z* (i.e., check if z divides both numbers without remainder). If so, exit the loop.
   ii. Otherwise, decrement **z** by 1 and repeat the loop.
e. After exiting the loop, set ***z*** as the GCD and print its value with a suitable message.

Question 3.
Calculate sales bonus based on performance.

a. Define a list "***employees***", which represents the names of employees in a company. For example:
```
employees = ['Alice', 'Brian', 'Clara', 'David', 'Ella',
'Fred', 'Grace']
```
Print employees.

b. Define three lists ***Sales1***, ***Sales2***, and ***Sales3***, each containing positive integers representing sales figures (in CAD dollars) for three consecutive months. The number of elements in each list should be the same as the number of employees. For example:
```
Sales1 = [1500, 2000, 1800, 2200, 1700, 1600, 2100]
Sales2 = [1600, 2100, 1750, 2300, 1800, 1700, 2200]
Sales3 = [1700, 2200, 1900, 2400, 1850, 1800, 2300]
```
Print ***Sales1***, ***Sales2***, ***Sales3***.

c. The overall bonus for each employee is calculated based on the average sales over the three months:
   i. If the average monthly sales are at least 2000, the employee gets a bonus of $500.
   ii. Otherwise, the bonus is $200.

d. In a loop:
   i. Calculate the average sales for each employee.
   ii. Print a message including the employee's name, average sales, and bonus amount.

Submission:

1. You will earn a maximum of 100 points (accounts for 7%) for submitting your **source code** (*lastname_sol_A3.py)* within the specified deadline.
2. **Do not upload your source code to Brightspace as a <u>single zip/rar file</u>.** Such submissions will not be accepted and a mark of **zero** will be assigned.
3. Marks will be **deducted** if ***comments*** **and/or a *suitable message*[1]** are missing.
4. **Late submissions** will incur a penalty of 10% per day for **up to two days**. Submissions made beyond two days will not be accepted, and a mark of **zero** will be assigned.
5. **Unlimited resubmissions** are allowed. However, only the last submission will be considered. Note that if you resubmit after the deadline, a penalty will apply, even if an earlier version was submitted on time.

[1]**NOTE:** In this course, the phrase '*suitable message*' refers to some kind of description that describes what is being printed. For example, just printing the value **22** by itself does not tell the user anything useful; however, printing **'x = 22'** would add some context to what is being printed and so this would qualify as a '*suitable message*' if the value of x is to be printed.

Academic Integrity.

Plagiarism is a serious offense and can result in severe consequences such as receiving a grade of **zero** on the assignment/lab or even being asked to leave the program.

Copying or using someone else's code is considered plagiarism. This includes using code from online sources, previous labs/assignments, or from other students. Even if you have modified the code, our antiplagiarism software can still show it as plagiarism.

To avoid plagiarism, make sure to always use your own words and ideas when writing source code. Additionally, always check with your instructor to make sure you understand what is allowed and what is not in terms of using outside resources.

Remember that academic integrity is essential for your own personal and professional growth, and it is your responsibility to uphold these principles. So please take it seriously and always produce your own original work.